

Homological Algebra 1

Labix

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Abstract

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1 Algebra of Chain Complexes

1.1 Chain Complexes

We begin with a very important notion in homological algebra. A chain complex records a sequence of abelian groups together with group homomorphisms that connect them up. All of homology and cohomology starts with establish a chain complex out of a topological space.

Definition 1.1.1 (Chain Complex) Let R be a ring. A chain complex $(C_\bullet, \partial_\bullet)$ is a family of R -modules C_n for $n \in \mathbb{Z}$ and R -module homomorphisms $\partial_n : C_n \rightarrow C_{n-1}$ such that $\partial_n \circ \partial_{n+1} = 0$ for all n . Explicitly, we have a diagram as follows:

$$\cdots \longrightarrow C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \longrightarrow \cdots$$

for which we require that

$$\text{im}(\partial_{n+1}) \subseteq \ker(\partial_n)$$

for each n .

We can relate short exact sequences with chain complexes. In particular, one can always extract short exact sequences from chain complexes.

Lemma 1.1.2

Let R be a ring. Let $(C_\bullet, \partial_\bullet)$ be a chain complex over R . Then for any n , the sequence

$$0 \longrightarrow Z_n(C_\bullet) \xrightarrow{\iota} C_n \xrightarrow{\partial_n} B_{n-1}(C_\bullet) \longrightarrow 0$$

is a short exact sequence. Moreover, if the terms of the chain complex consist of free abelian groups, then the sequence is split exact.

Proof

The inclusion map is injective. ∂_n is also surjective on its image. Finally, $\ker(\partial_n)$ is by definition $Z_n(C_\bullet) \cong \text{im}(\iota)$. Now since subgroups of a free abelian group are free, we conclude that $B_{n-1} \leq C_n$ is free. Recall that if the third term of a short exact sequence is free, the sequence is split. Hence we conclude. $\blacksquare$

It is routine to also define maps between chain complexes. Indeed in Groups and Rings we introduced groups and group homomorphisms. In Topology we defined topological spaces and continuous maps. For chain complexes we have the notion of chain maps.

Definition 1.1.3 (Chain Map)

Let R be a ring. Let $(C_\bullet, \partial_\bullet)$ and $(D_\bullet, \partial'_\bullet)$ be two chain complexes over R . A chain map $f_\bullet : C_\bullet \rightarrow D_\bullet$ is a family of R -module homomorphisms

$$f_n : C_n \rightarrow D_n$$

such that $\partial'_n \circ f_n = f_{n-1} \circ \partial_n$ for all n . Explicitly, we have the following commutative diagram:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial_{n+1}} & C_n & \xrightarrow{\partial_n} & C_{n-1} \longrightarrow \cdots \\ & & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} \\ \cdots & \longrightarrow & D_{n+1} & \xrightarrow{\partial'_{n+1}} & D_n & \xrightarrow{\partial'_n} & D_{n-1} \longrightarrow \cdots \end{array}$$

It is customary to drop the $n \in \mathbb{N}$ in the notation as it is usually implicit. So for example the condition for chain map becomes $\partial' \circ f = f \circ \partial$.

Lemma 1.1.4

Let R be a ring. Let $C_\bullet, D_\bullet, E_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ and $g : D_\bullet \rightarrow E_\bullet$ be chain maps. Then the term-wise composition of f and g also gives a chain map $g \circ f : C_\bullet \rightarrow E_\bullet$.

Definition 1.1.5 (The Category of Chain Complexes)

Let R be a ring. Define the category of chain complexes

$$\mathbf{Ch}(R)$$

over R to consist of the following data.

- The objects are the chain complexes over R
- The morphisms are the chain maps
- Composition is given by the term-wise composition of maps on all levels.

Definition 1.1.6 (Homology Group) Let R be a ring. Let $(C_\bullet, \partial_\bullet)$ be a chain complex over R .

- Define the the R -submodule of n -cycles of the chain complex to be

$$Z_n(C_\bullet) = \ker(\partial_n)$$

- Define the R -submodule of n -boundaries of the chain complex to be

$$B_n(C_\bullet) = \operatorname{im}(\partial_{n+1})$$

- Define the n th homology of the chain complex to be

$$H_n(C_\bullet) = \frac{Z_n(C_\bullet)}{B_n(C_\bullet)} = \frac{\ker(\partial_n)}{\operatorname{im}(\partial_{n+1})}$$

Definition 1.1.7 (Induced Homomorphisms) Let R be a ring. Let $C_\bullet$ and $D_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ be a chain map. Define the induced map on homology

$$f_* : H_n(C_\bullet) \rightarrow H_n(D_\bullet)$$

by the formula $f_*([z]) = [f(z)]$.

Lemma 1.1.8 Let R be a ring. Let $C_\bullet$ and $C'_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow C'_\bullet$ be a chain map. Then the induced map on homology

$$f_* : H_n(C_\bullet) \rightarrow H_n(C'_\bullet)$$

is a well defined R -module homomorphism.

Proof For every map $f_n : C_n \rightarrow C'_n$, we can restrict the domain to cycles so that we obtain a map $f_n : Z_n(C_\bullet) \rightarrow C'_n$. Using the relation given between the boundary operator and the family of maps, we check that this map descends to a map in homology.

Firstly, $f_n(Z_n(C_\bullet)) \subseteq Z_n(C'_\bullet)$. Indeed let $x \in Z_n(C_\bullet)$. Then we have that

$$\partial'_n(f_n(x)) = f_{n-1}(\partial_n(x)) = f_{n-1}(0) = 0$$

which means that $f_n(x)$ lies in the kernel of ∂'_n . Now we have a map $f_n : Z_n(C_\bullet) \rightarrow Z_n(C'_\bullet)$. At the same time, f_n also restricts to a map $f_n : B_n(C_\bullet) \rightarrow B_n(C'_\bullet)$. Indeed if $b \in B_n(C_\bullet)$, then there exists some $c \in C_{n+1}$ such that $\partial_{n+1}(c) = b$. Applying f_n on both sides give

$$\begin{aligned} f_n(\partial_{n+1}(c)) &= f_n(b) \\ \partial'_{n+1}(f_{n+1}(c)) &= f_n(b) \end{aligned}$$

This means that $f_n(b)$ is the boundary of the element $f_{n+1}(c) \in C_{n+1}$, and so f_n restricts to a map of boundaries. Now $f_n : H_n(C_\bullet) \rightarrow H_n(C'_\bullet)$ is well defined. Indeed if $b_1, b_2 \in B_n(C_\bullet)$ lie in the same coset, then $b_1 - b_2 \in B_n(C_\bullet)$ so that $f_n(b_1 - b_2) \in B_n(C'_\bullet)$ so that $f_n(b_1)$ and $f_n(b_2)$ lie in the same coset. Thus f_n is well defined. ■

We then have functoriality of the induced map.

Proposition 1.1.9 Let R be a ring. Let $C_\bullet, D_\bullet, E_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ and $g : D_\bullet \rightarrow E_\bullet$ be chain maps. Then the induced map on the homology groups satisfy the properties:

- $g_* \circ f_* = (g \circ f)_*$
- $\text{id}_* = \text{id}_{H_n}$

Proof Firstly, we have that

$$\partial \circ g_n \circ f_n = g_{n-1} \circ \partial \circ f_n = g_{n-1} \circ f_{n-1} \circ \partial$$

so that $g \circ f$ is indeed a chain map.

We have that $g_*(f_*([z])) = g_*([f(z)]) = [g(f(z))] = (g_* \circ f_*)([z])$. Also, we have that

$$\text{id}_*([z]) = [\text{id}(z)] = [z] = \text{id}_{H^n}([z])$$

and so we conclude. ■

Definition 1.1.10 (The Homology Functor) Let R be a ring. Define the homology functors

$$H_n(-) : \mathbf{Ch}(R) \rightarrow \mathbf{Ab}$$

for each n as follows.

- For each chain complex $C_\bullet$ over R , $H_n(C_\bullet)$ is the n th homology group of $C_\bullet$.
- For each chain map $f : C_\bullet \rightarrow C'_\bullet$,

$$H_n(f) : H_n(C_\bullet) \rightarrow H_n(C'_\bullet)$$

is the induced map on homology.

Definition 1.1.11 (Quasi-Isomorphisms)

Let R be a ring. Let $C_\bullet$ and $D_\bullet$ be a chain complex over R . Let $f : C_\bullet \rightarrow D_\bullet$ be a chain map. We say that f is a quasi-isomorphism if $H_n(f)$ is an isomorphism for all $n \in \mathbb{Z}$.

1.2 Chain Homotopies

In algebraic topology 1, we defined the notion of homotopies between two maps. In general, algebraic topologists are not only interested in maps between two objects such as groups or vector spaces, we are also interested in the maps between the maps. Therefore homotopies play such a crucial role in Algebraic Topology 1. In order to produce results similar to homotopy invariance of the fundamental groups, we will need the notion of homotopies between chain maps.

One can think of the chain complexes as our objects, chain maps as morphisms between chain complexes and morphisms between chain maps as chain homotopies. This is precisely the notion of the category of chain complexes.

Definition 1.2.1 (Chain Homotopy) Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ and $g : C_\bullet \rightarrow D_\bullet$ be chain maps. Then a chain homotopy from f to g is a collection of morphisms

$$\eta_n : C_n \rightarrow D_{n+1} \quad \text{such that} \quad g_n - f_n = \partial'_{n+1}\eta_n + \eta_{n-1}\partial_n$$

for all $n \in \mathbb{Z}$. Explicitly, the following diagram is commutative:

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial_{n+1}} & C_n & \xrightarrow{\partial_n} & C_{n-1} \longrightarrow \cdots \\
& \searrow g-f & \downarrow & \swarrow \eta_n & \downarrow g-f & \swarrow \eta_{n-1} & \downarrow g-f \\
\cdots & \longrightarrow & D_{n+1} & \xrightarrow{\partial'_{n+1}} & D_n & \xrightarrow{\partial'_n} & D_{n-1} \longrightarrow \cdots
\end{array}$$

In this case, f and g are said to be chain homotopic and we write $f \simeq g$.

Lemma 1.2.2 Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Then the relation $\simeq$ on the chain maps from $C_\bullet$ to $D_\bullet$ is an equivalence relation.

Proof Let $f_\bullet, g_\bullet, h_\bullet : C_\bullet \rightarrow D_\bullet$ be chain maps. Then it is clear that $f_\bullet \simeq f_\bullet$ by the 0 map. Indeed we have that $\partial 0 + 0 \partial = 0 = f - f$. Thus $\simeq$ is reflexive. if $f_\bullet$ and $g_\bullet$ are chain homotopic by η , then we have the identity

$$\partial \eta + \eta \partial = g - f$$

But then we have

$$\begin{aligned}
f - g &= -\partial \eta - \eta \partial \\
&= \partial(-\eta) + (-\eta)\partial
\end{aligned}$$

which means that $-\eta$ gives a chain homotopy from g to f .

Now if $g_\bullet$ and $h_\bullet$ are chain homotopic by ν , then we have that

$$\begin{aligned}
h_\bullet - f_\bullet &= h_\bullet - g_\bullet + g_\bullet - f_\bullet \\
&= \partial \nu + \nu \partial + \partial \eta + \eta \partial \\
&= \partial(\nu + \eta) + (\nu + \eta)\partial
\end{aligned}$$

so that $\nu + \eta$ defines a chain homotopy from f to h . ■

One consequence of chain homotopies is the following.

Lemma 1.2.3 Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ and $g : C_\bullet \rightarrow D_\bullet$ be chain maps. If f and g are chain homotopic, then f and g induce the same map

$$f_n = g_n : H_n(X) \rightarrow H_n(Y)$$

in homology.

Proof Let $c \in \ker(\partial_n)$ be an n -cycle. Using the equation for chain homotopy, we have that

$$\begin{aligned}
b(c) - a(c) &= \partial'_{n+1}(\eta_n(c)) + \eta_{n-1}(\partial(c)) \\
&= \partial'_{n+1}(\eta(c))
\end{aligned}$$

is a boundary in $\text{im}(\partial'_{n+1}) \subseteq C'_n$. Thus $b_n(c)$ and $a_n(c)$ are of the same coset in $H_n(X)$. ■

Chain homotopies are also well defined in compositions.

Proposition 1.2.4 Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Let $f_1, g_1 : C_\bullet \rightarrow D_\bullet$ and $f_2, g_2 : C_\bullet \rightarrow D_\bullet$ be chain maps. If $f_1 \simeq g_1$ and $f_2 \simeq g_2$, then $f_2 \circ f_1 \simeq g_2 \circ g_1$.

Proof The chain homotopies between f_1 and g_1 imposes the identity

$$\partial \eta + \eta \partial = g_1 - f_1$$

for $\eta : C_\bullet \rightarrow D_\bullet$ the given chain homotopy. Similarly, for $\nu : D_\bullet \rightarrow E_\bullet$ we have the identity

$$\partial \nu + \nu \partial = g_2 - f_2$$

Then we have that

$$\begin{aligned}
 g_2 \circ g_1 - f_2 \circ f - 1 &= g_2 \circ g_1 - g_2 \circ f_1 + g_2 \circ f_1 - f_2 \circ f_1 \\
 &= g_2(g_1 - f_1) + (g_2 - f_2) \circ f_1 \\
 &= g_2(\partial\eta + \eta\partial) + (\partial\nu + \nu\partial) \circ f_1 \\
 &= \partial g_2\eta + g_2\eta\partial + \partial\nu f_1 + \nu f_1\partial \\
 &= \partial(g_2\eta + \nu f_1) + (g_2\eta + \nu f_1)\partial
 \end{aligned}$$

Thus $g_2\eta + \nu f_1 : C_n \rightarrow E_{n+1}$ would be a valid chain homotopy from $g_2 \circ g_1$ to $f_2 \circ f_1$. ■

While chain maps induce a map on the homology groups, we will see that chain homotopy equivalences induces isomorphisms on the homology groups, intuitively because they produce an inverse for each map on the level of homology.

Definition 1.2.5 (Chain Homotopy Equivalence) Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . We say that they are chain homotopy equivalence if there exists chain maps $f_\bullet : C_\bullet \rightarrow D_\bullet$ and $g_\bullet : D_\bullet \rightarrow C_\bullet$ such that there are chain homotopies

$$g_\bullet \circ f_\bullet \simeq \text{id}_{C_\bullet} \quad \text{and} \quad f_\bullet \circ g_\bullet \simeq \text{id}_{D_\bullet}.$$

In this case, we say that g is a chain homotopy inverse of f .

Lemma 1.2.6 Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Suppose that $f : C_\bullet \rightarrow D_\bullet$ is a chain map that induces a chain homotopy equivalence between $C_\bullet$ and $D_\bullet$. Then f induces an isomorphism

$$H_n(C_\bullet) \cong H_n(D_\bullet)$$

in all degrees $n \in \mathbb{N}$.

Proof Suppose that g is the chain homotopy inverse of f . We know that $g_\bullet \circ f_\bullet \simeq \text{id}_{C_\bullet}$ which means that they induce the same map:

$$g_* \circ f_* = \text{id}_{H_n(C_\bullet)}$$

Similarly the chain homotopies $f_\bullet \circ g_\bullet \simeq \text{id}_{D_\bullet}$ induce the same map

$$f_* \circ g_* = \text{id}_{H_n(D_\bullet)}$$

as the identity. Then these two identities mean that f_* is both injective and surjective. ■

Similar to homotopies in Algebraic Topology, chain homotopy also defines an equivalence relation on all the chain maps.

Proposition 1.2.7 Chain homotopy equivalence defines an equivalence relation on all chain complexes.

Proof Clearly any chain complex is chain homotopy equivalent to itself by the identity map. If $C_\bullet$ and $D_\bullet$ are chain homotopy equivalent by the chain maps $a_\bullet : C_\bullet \rightarrow D_\bullet$ and $b_\bullet : D_\bullet \rightarrow C_\bullet$, then we have the identities $b_\bullet \circ a_\bullet = \text{id}_{C_\bullet}$ and $a_\bullet \circ b_\bullet = \text{id}_{D_\bullet}$. We can then read them in reverse so that $D_\bullet$ and $C_\bullet$ are chain homotopy equivalence by the maps $b_\bullet$ and $a_\bullet$.

Suppose further that $D_\bullet$ and $E_\bullet$ are chain homotopy equivalent via the maps $u_\bullet : D_\bullet \rightarrow E_\bullet$ and $v_\bullet : E_\bullet \rightarrow D_\bullet$. Then the maps $u_\bullet \circ a_\bullet$ and $b_\bullet \circ v_\bullet$ give a chain homotopy equivalence between $C_\bullet$ and $E_\bullet$. Indeed, upon composition, we have that they are chain homotopic to the identity maps. ■

Be careful that chain homotopy defines an equivalence relation on all chain maps between two fixed chain complexes, while chain homotopy equivalence defines an equivalence relation on all chain com-

plexes. One is an equivalence relation on the objects and one is on the morphisms.

This is reminiscent to that of ordinary homotopies between continuous maps. Namely, homotopies defines an equivalence relation on all continuous maps between two fixed topological spaces, while homotopy equivalence defines an equivalence relation on all topological spaces.

1.3 Sequences of Chain Complexes

One can even define short exact sequences of chain complexes themselves.

Definition 1.3.1 (Short Exact Sequence of Chain Complexes) Let R be a ring. Let the following be a sequence of homomorphisms in $\mathbf{Ch}(R\mathbf{Mod})$.

$$0 \longrightarrow A_\bullet \xrightarrow{f} B_\bullet \xrightarrow{g} C_\bullet \longrightarrow 0$$

We say that it is short exact if for each $n \in \mathbb{N}$, the following sequence

$$0 \longrightarrow A_n \xrightarrow{f_n} B_n \xrightarrow{g_n} C_n \longrightarrow 0$$

is short exact.

Explicitly, we can see that a short exact sequence of chain complexes can be written in diagrams as:

$$\begin{array}{ccccccc} & \vdots & & \vdots & & \vdots & \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & A_{n+1} & \xrightarrow{f_{n+1}} & B_{n+1} & \xrightarrow{g_{n+1}} & C_{n+1} \longrightarrow 0 \\ & & \partial \downarrow & & \partial \downarrow & & \partial \downarrow \\ 0 & \longrightarrow & A_n & \xrightarrow{f_n} & B_n & \xrightarrow{g_n} & C_n \longrightarrow 0 \\ & & \partial \downarrow & & \partial \downarrow & & \partial \downarrow \\ 0 & \longrightarrow & A_{n-1} & \xrightarrow{f_{n-1}} & B_{n-1} & \xrightarrow{g_{n-1}} & C_{n-1} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & \vdots & & \vdots & & \vdots \end{array}$$

Theorem 1.3.2 Let R be a ring. Let the following be a short exact sequence of chain complexes in $\mathbf{Ch}(R\mathbf{Mod})$.

$$0 \longrightarrow A_\bullet \xrightarrow{f} B_\bullet \xrightarrow{g} C_\bullet \longrightarrow 0$$

Then there exists a connecting homomorphism $d : H_n(C_\bullet) \rightarrow H_{n-1}(A_\bullet)$ such that the following sequence of homology groups

$$\cdots \longrightarrow H_{n+1}(C_\bullet) \xrightarrow{d} H_n(A_\bullet) \xrightarrow{(f_n)_*} H_n(B_\bullet) \xrightarrow{(g_n)_*} H_n(C_\bullet) \xrightarrow{d} H_{n-1}(A_\bullet) \longrightarrow \cdots$$

is long exact. Explicitly, d is defined as follows.

- Let $[c] \in H_n(C_\bullet)$. By surjectivity of g_n there exists $b \in B_n$ such that $g_n(b) = c$.
- We have $c \in Z_n(C_\bullet)$. Because $g_{n-1}(\partial_B(b)) = \partial_C(g_n(b)) = \partial_C(c) = 0$, we have that $\partial_B(b) \in \ker(g_{n-1}) = \operatorname{im}(f_{n-1})$. So there exists $a \in A_{n-1}$ such that $f_{n-1}(a) = \partial_B(b)$.

We then define

$$d([c]) = [a]$$

Proof We want to show that this is well defined. We have made two choices in the definition of d .

- Suppose that $b, b' \in B_n$ are such that $g_n(b) = c = g_n(b')$. Then $b - b' \in \ker(g_n) = \operatorname{im}(f_n)$ means that there exists $a' \in A_n$ such that $f_n(a') = b - b'$. Using the same argument from the construction of the map d , we obtain $a, a' \in A_{n-1}$ such that $f_{n-1}(a) = \partial_B(b)$ and $f_{n-1}(a') = \partial_B(b')$. This means that $f_{n-1}(a - a') = \partial_B(b - b') = \partial_B(f_n(a'))$. By commutativity of the diagram, we have that

$$\partial_B(f_n(a')) = f_{n-1}(\partial_A(a'))$$

so that we have $f_{n-1}(a - a') = f_{n-1}(\partial_A(a''))$. Since f_{n-1} is injective, we conclude that $a - a' = \partial_A(a'')$. But this means that $[a] = [a']$. This implies that the choice of b did not matter in the definition of d .

- Suppose that $a, a' \in A_{n-1}$ such that $f_{n-1}(a) = \partial_B(b) = f_{n-1}(a')$. Then by injectivity of f_{n-1} we conclude that $a = a'$.

We now need to show that d is well defined under equivalence classes. Suppose that $c \in \text{im}(\partial_C)$. This implies that there exists $c' \in C_{n+1}$ such that $\partial_C(c') = c$. By surjectivity of g_{n+1} , there exists $b \in B_{n+1}$ such that $g_{n+1}(b) = c'$. By commutativity of the diagram, we have that

$$c = \partial_C(g_{n+1}(b)) = g_n(\partial_B(b))$$

Following the construction of d , we obtain $a \in A_{n-1}$ such that $f_{n-1}(a) = \partial_B(\partial_B(b)) = 0$. Hence $a = 0$. We conclude that d is a well defined function. We next show that d is an R -module homomorphism.

Suppose that $[c], [c'] \in H_n(C_\bullet)$. Following the construction of d we obtain $b, b' \in B_n$ and $a, a' \in A_{n-1}$ such that

$$g_n(b) = c \quad g_n(b') = c' \quad f_{n-1}(a) = \partial_B(b) \quad f_{n-1}(a') = \partial_B(b')$$

We notice that $g_n(b + b') = c + c'$ and $f_{n-1}(a + a') = \partial_B(b + b')$ so that

$$d([c + c']) = [a + a'] = [a] + [a'] = d([c]) + d([c'])$$

The prove is similar in showing that $d([rc]) = rd([c])$

We also need to show that the long sequence is exact. ■

Notice that the above theorem is essentially an application of the snake lemma. One can conclude immediately by the snake lemma.

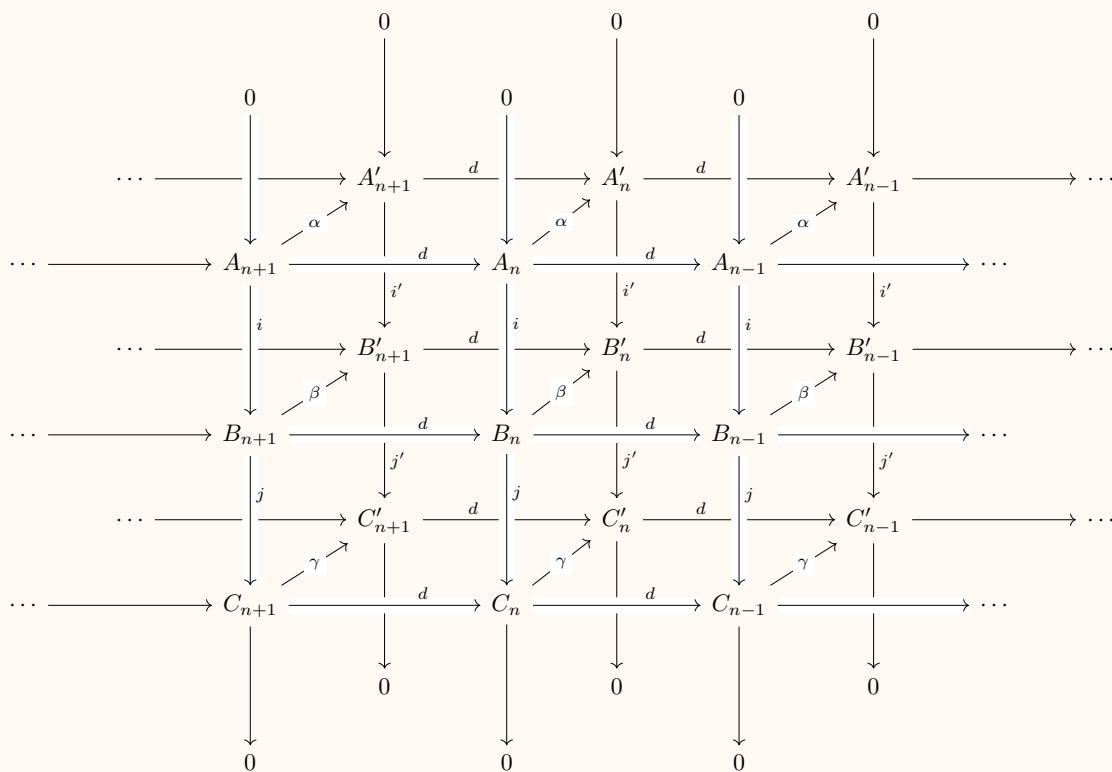
The following theorem looks horrifying but it is not as terrible as it looks. In essence, we want to show a naturality condition. This means that given a morphism of short exact sequences of chain complexes, we want an induced map that satisfies some commutative square.

Theorem 1.3.3 Let R be a ring. Let the following be a short exact sequence of chain complexes in $\text{Ch}(R\text{Mod})$. Let $A_\bullet, B_\bullet, C_\bullet, A'_\bullet, B'_\bullet, C'_\bullet$ be six chain complexes and let the following

$$0 \longrightarrow A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{j} C_\bullet \longrightarrow 0$$

$$0 \longrightarrow A'_\bullet \xrightarrow{i'} B'_\bullet \xrightarrow{j'} C'_\bullet \longrightarrow 0$$

be two short exact sequence of chain complexes. Let the following diagram be a morphism of the two short exact sequence of chain complexes.



Then the induced diagram

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & H_n(A) & \xrightarrow{i_*} & H_n(B) & \xrightarrow{j_*} & H_n(C) \xrightarrow{\partial} H_{n-1}(A) \longrightarrow \cdots \\
 & & \downarrow \alpha_* & & \downarrow \beta_* & & \downarrow \gamma_* \\
 \cdots & \longrightarrow & H_n(A') & \xrightarrow{i'_*} & H_n(B') & \xrightarrow{j'_*} & H_n(C') \xrightarrow{\partial} H_{n-1}(A') \longrightarrow \cdots
 \end{array}$$

is a commutative diagram.

Proof The first square is commutative since $\beta \circ i = i' \circ \alpha$ and i, i', α, β are all chain maps so that $\beta_* \circ i_* = i'_* \circ \alpha_*$ by proposition 1.1.5. Similarly, the second square is also commutative since $\gamma \circ j = j' \circ \beta$ implies $\gamma_* \circ j_* = j'_* \circ \beta_*$. Now recall that the connecting homomorphism is defined by $\partial([c]) = [a]$ where $j(b) = c$ and $i(a) = d(b)$. Since

$$\gamma(c) = \gamma(j(b)) = j'(\beta(b))$$

and

$$i'(\alpha(a)) = \beta(i(a)) = \beta(d(b)) = d(\beta(b))$$

we have that $\partial[\gamma(c)] = [\alpha(a)]$. Thus we have that

$$\partial(\gamma_*([c])) = \alpha_*([a]) = \alpha_*(\partial([c]))$$

and so we conclude. ■

This naturality condition is also satisfied by the exact sequence in the snake lemma.

1.4 Cochain Complexes

Definition 1.4.1 (Cochain Complexes) Let R be a ring. A cochain complex $(C^\bullet, \partial^\bullet)$ over R is a family of R -modules C^n for $n \in \mathbb{Z}$ and R -module homomorphisms $\delta^n : C^{n-1} \rightarrow C^n$ such that $\delta^{n+1} \circ \delta^n = 0$ for all n . In other words, we have the diagram:

$$\cdots \longleftarrow C^{n+1} \xleftarrow{\delta^{n+1}} C^n \xleftarrow{\delta^n} C^{n-1} \longleftarrow \cdots$$

Notice that algebraically, there is no difference between a chain complex and a cochain complex, other

than the fact that the boundary maps run in the other direction. For $(C_\bullet, \partial_\bullet)$ a chain complex, we can form a cochain complex by setting $C^n = C_{-n}$ and then using the same boundary maps.

Definition 1.4.2 (Category of Cochain Complexes) Let R be a ring. Define the category of cochain complexes

$$\mathbf{CCh}(R)$$

to consist of the following data.

- The objects are the cochain complexes over R
- The morphisms are the chain maps
- Composition is given by the composition of maps on all levels of the chain map.

Because cochain complexes is the same data as a chain complex, we can also define homology groups for cochain complexes. We call them the cohomology groups of the cochain complex.

Definition 1.4.3 (Cohomology Groups) Let R be a ring. Let $(C^\bullet, \partial^\bullet)$ be a cochain complex over R . Define the n th cohomology group of the cochain complex to be

$$H^n(C^\bullet, \partial^\bullet) = \frac{\ker(\partial^{n+1} : C^n \rightarrow C^{n+1})}{\operatorname{im}(\partial^n : C^{n-1} \rightarrow C^n)} = H_n(C^\bullet, \partial^\bullet)$$

for $n \in \mathbb{N}$.

The construction of cohomology groups are functorial because the homology groups of a chain complex also are.

Definition 1.4.4 (The Cohomology Functor) Let R be a ring. Let $n \in \mathbb{N}$. Define the n th cohomology functor to be the functor

$$H^n : \mathbf{CCh}(R) \rightarrow \mathbf{Ab}$$

to consist of the following data.

- For each cochain complex $(C^\bullet, \delta^\bullet)$ over R , define $H^n(C^\bullet, \delta^\bullet)$ to be the n th cohomology group of the cochain complex.
- For each chain map $f^\bullet : (C^\bullet, \delta^\bullet) \rightarrow (D^\bullet, \delta^\bullet)$, the induced map

$$f^* : H^n(C^\bullet, \delta^\bullet) \rightarrow H^n(D^\bullet, \delta^\bullet)$$

is defined by $[z] \mapsto [f^n(z)]$.

Notice that obtaining cohomology from chain complexes is covariant.

1.5 Producing a Cochain Complex from a Chain Complex

Let R be a ring. Let M, N be R -modules. Recall that $\operatorname{Hom}_R(M, N)$ is only an abelian group, and at most a $Z(R)$ -module in general. However, when R is commutative, $\operatorname{Hom}_R(M, N)$ becomes an R -module. Here we treat the more general case and consider the case when R is a general ring.

Definition 1.5.1 (Dualization of a Chain Complex) Let R be a commutative ring. Let $(C_\bullet, \partial_\bullet)$ be a chain complex. Let M be an R -module. For each R -module C_n , define the associated cochain group to be

$$C^n = \operatorname{Hom}_R(C_n, M)$$

Also define the coboundary map $\delta^n = \partial_n^* : C^{n-1} \rightarrow C^n$ by

$$\delta^n(\phi)(\alpha) = \phi(\partial_{n+1}(\alpha))$$

Lemma 1.5.2 Let $(C_\bullet, \partial_\bullet)$ be a chain complex and let $C^\bullet$ and $\delta^\bullet$ be the associated cochain groups and coboundary maps respectively. Then

$$\delta^n \circ \delta^{n-1} = 0$$

for all n such that $(C^\bullet, \delta^\bullet)$ is a cochain complex.

Definition 1.5.3 (The Hom Functor on Chain Complexes) Let R be a ring. Let M be an R -module. Define the (contravariant) Hom functor on chain complexes to be the functor

$$\mathrm{Hom}_R(-; M) : \mathbf{Ch}(R) \rightarrow \mathbf{CCh}(\mathbb{Z})$$

consisting of the following data.

- For $(C_\bullet, \partial_\bullet)$ a chain complex, define a cochain complex by $C^n = \mathrm{Hom}_R(C_n, M)$ and coboundary map by $\delta_n = \partial_n^*$
- For a chain map $f_\bullet : (C_\bullet, \delta_\bullet) \rightarrow (D_\bullet, \delta_\bullet)$, define a chain map $f^* : D^\bullet \rightarrow C^\bullet$ by $\varphi \mapsto \varphi \circ f$.

If R is a commutative ring and M, N are R -modules, then $\mathrm{Hom}_R(M, N)$ can be given the structure of an R -module. Therefore the above Hom functor upgrades into a functor

$$\mathrm{Hom}_R(-; M) : \mathbf{Ch}(R) \rightarrow \mathbf{CCh}(R)$$

Proposition 1.5.4 Let $C_\bullet, D_\bullet, E_\bullet \in \mathbf{Ch}(R)$ be chain complexes such that the short exact sequence

$$0 \longrightarrow C_\bullet \xrightarrow{f} D_\bullet \xrightarrow{g} E_\bullet \longrightarrow 0$$

of chain complexes is degree wise split (this means that $D_n \cong C_n \oplus E_n$ for all $n \in \mathbb{N}$). Then the dual of the chain complex

$$0 \longrightarrow E^\bullet \xrightarrow{g^*} D^\bullet \xrightarrow{f^*} C^\bullet \longrightarrow 0$$

is also a short exact sequence of chain complexes.

We have now defined a series of functors. We also recap on the material in section 1.

- We have defined the homology functor of chain complexes

$$H_n : \mathbf{Ch}(R) \rightarrow \mathbf{Mod}_R$$

for each $n \in \mathbb{N}$.

- Using the very same functor, but thinking about cochain complexes as chain complexes in the wrong direction, we have also defined the cohomology functor of cochain complexes

$$H^n : \mathbf{CCh}(R) \rightarrow \mathbf{Mod}_R$$

for each $n \in \mathbb{N}$.

- We have a functorial method of dualizing chain complexes into cochain complexes by the functor

$$\mathrm{Hom}_R(-, M) : \mathbf{Ch}(R) \rightarrow \mathbf{CCh}(R)$$

for any R -module M .

1.6 Koszul Complexes

Definition 1.6.1 (Koszul Complex)

Let R be a ring. Let M be a left R -module. Let $\varphi : M \rightarrow R$ be an R -module homomorphism. Define the Koszul complex $K_\bullet(\varphi)$ of φ to be the chain complex given as follows.

- For each $n \in \mathbb{N}$, define

$$K_n(\varphi) = \bigwedge_{i=1}^n M$$

and $K_n(\varphi) = 0$ for $n < 0$.

- Define the differential $d : K_n(M) \rightarrow K_{n-1}(M)$ by

$$d(m_1 \wedge \cdots \wedge m_n) = \sum_{i=1}^n (-1)^{i+1} \varphi(m_i) m_1 \wedge \cdots \wedge \widehat{m_i} \wedge \cdots \wedge m_n$$

for $m_1, \dots, m_n \in M$ and then extend it R -linearly.

When M is a finitely generated free R -module, it suffices to see where a set of R -basis is mapped to.

Definition 1.6.2 (Koszul Complex on Central Elements)

Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be an R -module. Define the Koszul complex of M on $x_1, \dots, x_n$ to be

$$K_\bullet(x_1, \dots, x_n; M) = K_\bullet(\varphi)$$

where $\varphi : M^n \rightarrow M$ is the map $\varphi(m_1, \dots, m_n) = \sum_{i=1}^n x_i \cdot m_i$.

Let R be a ring. Let $x_1, \dots, x_n \in R$ be central elements. Let $e_1, \dots, e_n$ be an R -basis for R^n . Then we may describe the differential on the R -basis of $\Lambda^k(R^n)$ as follows:

$$d(e_{i_1} \wedge \dots \wedge e_{i_k}) = \sum_{j=1}^k (-1)^{j+1} x_j e_{i_1} \wedge \dots \wedge \widehat{e_{i_j}} \wedge \dots \wedge e_{i_k}$$

Example 1.6.3

Let R be a ring. Let $x \in Z(R)$. Then the Koszul complex $K_\bullet(x)$ is the chain complex:

$$0 \longrightarrow R \xrightarrow{x} R \longrightarrow 0$$

Lemma 1.6.4

Let R be a ring. Let $x, y \in Z(R)$ be central elements. Then we have

$$K_\bullet(x, y) = \text{Tot}^\oplus(K_\bullet(x) \otimes_R K_\bullet(y))$$

Lemma 1.6.5

Let R be a commutative ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let $A = (a_{ij}) \in GL_n(R)$ be an invertible matrix. Then there is an isomorphism

$$K_\bullet(x_1, \dots, x_n) \cong K_\bullet\left(\sum_{i=1}^n a_{1i} f_i, \dots, \sum_{i=1}^n a_{ni} f_i\right)$$

For an example, let R be a commutative ring. Let $x, y \in R$. Then the Koszul complex $K(x, y)$ is given by

$$0 \longrightarrow R^2 \wedge R^2 \longrightarrow R^2 \longrightarrow R \longrightarrow 0$$

The differentials are given as follows.

- The first differential $R^2 \rightarrow R$ is given by $(r, s) \mapsto rx + sy$. It can also be given as a 1×2 matrix as $\begin{pmatrix} x & y \end{pmatrix}$. Also alternatively, we can write an R -basis for R^2 with $(1, 0)$ and $(0, 1)$. Then define the map $R^2 \rightarrow R$ by $(1, 0) \mapsto x$ and $(0, 1) \mapsto y$.
- Now recall that $R^2 \wedge R^2$ has R -basis given by $(1, 0) \wedge (0, 1)$ and so there is an isomorphism $R^2 \wedge R^2 \cong R$. Then the second differential $R \rightarrow R^2$ is given by $1 \mapsto (x, -y)$.

Definition 1.6.6 (Koszul (Co)Homology) Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be an R -module. Define the Koszul homology of M with respect to $x_1, \dots, x_n$ by

$$H_k^{\text{Kos}}(x_1, \dots, x_n; M) = H_k(K(x_1, \dots, x_n; M))$$

Define the Koszul cohomology of M with respect to the central elements by

$$H_{\text{Kos}}^k(x_1, \dots, x_n; M) = H^k(\text{Hom}_R(K(x_1, \dots, x_n)), M)$$

Lemma 1.6.7

Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be an R -module. Then the following are true.

- $H_0^{\text{Kos}}(x_1, \dots, x_n; M) = \frac{M}{(x_1, \dots, x_n)M}$
- $H_n^{\text{Kos}}(x_1, \dots, x_n; M) = \{m \in M \mid x_i m = 0 \text{ for all } i\}$

Proof Notice that the map $K_1(x_1, \dots, x_n; M) \rightarrow K_0(x_1, \dots, x_n; M)$ is given by $(m_1, \dots, m_n) \mapsto \sum_{i=1}^n x_i \cdot m_i$. Hence the image of the map is given by $(x_1, \dots, x_n)M$. Thus we have

$$H_0^{\text{Kos}}(x_1, \dots, x_n; M) = \frac{\ker(M \rightarrow 0)}{\text{im}(K_1(x_1, \dots, x_n; M) \rightarrow K_0(x_1, \dots, x_n; M))} = \frac{M}{(x_1, \dots, x_n)M}$$

Similarly, the map $K_n(x_1, \dots, x_n; M) \rightarrow K_{n-1}(x_1, \dots, x_n; M)$ is given by

$$m \mapsto \begin{pmatrix} (-1)^{n+1} x_n \\ \vdots \\ (-1)^{i+1} x_i \\ \vdots \\ x_1 \end{pmatrix}$$

Hence the kernel of the map is given by $\{m \in M \mid x_i m = 0 \text{ for all } 1 \leq i \leq n\}$. Thus we have

$$H_n^{\text{Kos}}(x_1, \dots, x_n; M) = \frac{\ker(K_n(x_1, \dots, x_n; M) \rightarrow K_{n-1}(x_1, \dots, x_n; M))}{\text{im}(0 \rightarrow M)} = \{m \in M \mid x_i m = 0 \text{ for all } i\}$$

So we are done. ■

Theorem 1.6.8 (Kunneth Theorem) Let R be a ring. Let $x_1, \dots, x_n \in R$ be central elements. Let $C_\bullet$ be a chain complex of R -modules. Then there is an exact sequence given by

$$0 \longrightarrow H_0(x_1, \dots, x_n; H_q(C_\bullet)) \longrightarrow H_q^{\text{Tot}}(K(x_1, \dots, x_n) \otimes_R C_\bullet) \longrightarrow H_1(x_1, \dots, x_n; H_{q-1}(C_\bullet)) \longrightarrow 0$$

Theorem 1.6.9 (Self Duality of the Koszul Complex)

Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be a left R -module. Then there is a natural isomorphism of chain complexes

$$\text{Hom}(K(x_1, \dots, x_n); M) \cong K(x_1, \dots, x_n; M)$$

where we think of the left complex as a chain complex.

The self duality of the Koszul complex means that we could have equally defined the differential to be going up the chain instead of down. In this case, the complex can be defined as

$$0 \longrightarrow \bigwedge_{i=1}^n R^n \xrightarrow{d_n} \bigwedge_{i=1}^{n-1} R^n \longrightarrow \cdots \longrightarrow R^n \xrightarrow{d_1} R \longrightarrow 0$$

where the differential $d_k : \bigwedge_{i=1}^{k-1} R^n \rightarrow \bigwedge_{i=1}^k R^n$ is now defined by

$$d(e_{i_1} \wedge \cdots \wedge e_{i_{k-1}}) = \sum_{j=1}^n x_j e_j \wedge e_{i_1} \wedge \cdots \wedge e_{i_{k-1}}$$

Corollary 1.6.10 Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be a left R -module. Then there is an isomorphism

$$H_p^{\text{Kos}}(x_1, \dots, x_n; M) \cong H_{\text{Kos}}^{n-p}(x_1, \dots, x_n; M)$$

2 Ext and Tor Groups

2.1 Injective Modules

Definition 2.1.1 (Injective Modules)

Let R be a ring. Let I be an R -module. We say that I is injective if for all short exact sequences of R -modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

we have that

$$0 \longrightarrow \operatorname{Hom}_R(M_3, I) \longrightarrow \operatorname{Hom}_R(M_2, I) \longrightarrow \operatorname{Hom}_R(M_1, I) \longrightarrow 0$$

is an exact sequence of R -modules.

Proposition 2.1.2

Let R be a ring. Let I be an R -module. Then I is injective if and only if for R -modules M, N and all injective homomorphisms $f : M \rightarrow N$ and all maps $u : M \rightarrow I$, there exists a morphism $N \rightarrow I$ such that the following diagram commutes:

$$\begin{array}{ccc} M & \xrightarrow{u} & I \\ f \downarrow & \nearrow \exists! & \\ N & & \end{array}$$

Proof

Suppose that I is injective. Suppose that $f : M \rightarrow N$ is injective. Then we have an exact sequence

$$0 \rightarrow M \xrightarrow{f} N \rightarrow \operatorname{coker}(f) \rightarrow 0$$

and hence the exact sequence

$$0 \longrightarrow \operatorname{Hom}_R(\operatorname{coker}(f), I) \longrightarrow \operatorname{Hom}_R(N, I) \longrightarrow \operatorname{Hom}_R(M, I) \longrightarrow 0$$

Since the map $\operatorname{Hom}_R(N, I) \rightarrow \operatorname{Hom}_R(M, I)$ is surjective, we are done.

For the other implication, suppose that $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ is a short exact sequence. From Rings and Modules we already know that the sequence

$$0 \longrightarrow \operatorname{Hom}_R(M_3, I) \longrightarrow \operatorname{Hom}_R(M_2, I) \longrightarrow \operatorname{Hom}_R(M_1, I)$$

is exact. It remains to show that $\operatorname{Hom}_R(M_2 \rightarrow I) \rightarrow \operatorname{Hom}_R(M_1, I)$ is surjective. But this is true by assumption. ■

Example 2.1.3

Let k be a field. Then every k -module is injective.

Proof

Let V, W, X be k -vector spaces. Let $T : V \rightarrow W$ and $u : V \rightarrow X$ be k -linear maps such that T is injective. I claim that V is injective, so let us define the map $S : W \rightarrow X$. Let B be a basis of V . Then $T(B)$ is linearly independent. For each $e \in T(B)$, define $S(e) = u(e)$. Now every linearly independent set of vectors can be extended to a basis. Hence there exists $C \subseteq W$ such that $T(B) \cup C$ is a basis of W . For each $f \in C$, choose any element $x_f \in X$. Define $S(f) = x_f$. Then S is a valid k -linear map. ■

Proposition 2.1.4 (Baer's Criterion)

Let R be a ring. Let M be a left R -module. Then M is injective if and only if for every left ideal I of R and R -module homomorphism $\phi : I \rightarrow M$, there exists a map $\bar{\phi}$ such that the following diagram commutes:

$$\begin{array}{ccc} I & \xrightarrow{\phi} & M \\ \downarrow & \nearrow \exists \bar{\phi} & \\ R & & \end{array}$$

2.2 Projective Modules**Definition 2.2.1 (Projective Modules)**

Let R be a ring. Let P be an R -module. We say that P is projective if for all short exact sequences of R -modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

we have that

$$0 \longrightarrow \operatorname{Hom}_R(P, M_1) \longrightarrow \operatorname{Hom}_R(P, M_2) \longrightarrow \operatorname{Hom}_R(P, M_3) \longrightarrow 0$$

is an exact sequence of R -modules.

Lemma 2.2.2

Let R be a ring. Let F be a free R -module. Then F is projective.

Proposition 2.2.3

Let R be a ring. Let P be an R -module. Then the following are equivalent.

- P is projective.
- For all R -modules M, N and all surjective homomorphisms $f : M \rightarrow N$ and $u : P \rightarrow N$, there exists a morphism $P \rightarrow M$ such that the following diagram commutes:

$$\begin{array}{ccc} & & M \\ & \nearrow \exists! & \downarrow f \\ P & \xrightarrow{u} & N \end{array}$$

- $P \oplus Q$ is a free R -module for some R -module Q .

Proposition 2.2.4

Let R be a ring. Let P_i be an R -module for each $i \in I$. Then $\bigoplus_{i \in I} P_i$ is projective if and only if each P_i is projective.

2.3 Resolutions**Definition 2.3.1 (Resolution of Objects)**

Let R be a ring. Let M be an R -module.

- A left resolution of M is a chain complex $F_\bullet$ of R -modules together with a quasi-isomorphism $\iota : F_\bullet \rightarrow M$ depicted as

$$\begin{array}{ccccccc} \cdots & \longrightarrow & F_1 & \longrightarrow & F_0 & \longrightarrow & 0 \longrightarrow \cdots \\ & & \downarrow & & \downarrow \iota & & \downarrow \\ \cdots & \longrightarrow & 0 & \longrightarrow & M & \longrightarrow & 0 \longrightarrow \cdots \end{array}$$

- A right resolution of M is a cochain complex $F^\bullet$ of R -modules together with a quasi-isomorphism depicted as

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & 0 & \longrightarrow & M & \longrightarrow & 0 \longrightarrow \cdots \\
 & & \downarrow & & \downarrow \iota & & \downarrow \\
 \cdots & \longrightarrow & 0 & \longrightarrow & F^0 & \longrightarrow & F^1 \longrightarrow \cdots
 \end{array}$$

Definition 2.3.2 (Injective Resolutions)

Let R be a ring. Let M be an R -module. An injective resolution of M is a cochain complex $I^\bullet$ of R -modules such that the following are true.

- I^n is injective for all $n \geq 0$.
- $I^\bullet$ is a right resolution of M .

Example 2.3.3

An injective resolution of $\mathbb{Z}$ in $\mathbf{Ab}$ is given by

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \longrightarrow 0 \longrightarrow \cdots \\
 & & \downarrow & & \downarrow \text{incl.} & & \downarrow \\
 \cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Q} & \longrightarrow & \mathbb{Q}/\mathbb{Z} \longrightarrow 0 \longrightarrow \cdots
 \end{array}$$

Proposition 2.3.4

Let R be a ring. Let M, N be R -modules. Let $\phi : M \rightarrow N$ be an R -module homomorphism. Suppose that there are injective resolutions

$$\begin{array}{ccccccc}
 0 & \longrightarrow & M & \longrightarrow & I^0 & \longrightarrow & I^1 \longrightarrow \cdots \\
 & & \downarrow \phi & & & & \\
 0 & \longrightarrow & N & \longrightarrow & J^0 & \longrightarrow & J^1 \longrightarrow \cdots
 \end{array}$$

for M and N respectively. Then there exists a chain map extending ϕ such that the following diagram commutes:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & M & \longrightarrow & I^0 & \longrightarrow & I^1 \longrightarrow \cdots \\
 & & \downarrow \phi & & \downarrow \phi^0 & & \downarrow \phi^1 \\
 0 & \longrightarrow & N & \longrightarrow & J^0 & \longrightarrow & J^1 \longrightarrow \cdots
 \end{array}$$

Moreover, any two such chain maps are homotopic.

Corollary 2.3.5

Let R be a ring. Let M be an R -module. Let $I^\bullet$ and $J^\bullet$ be injective resolutions of M . Then $I^\bullet$ and $J^\bullet$ are chain homotopy equivalent.

Proposition 2.3.6

Let R be a ring. Let M be an R -module. Then M admits a unique injective resolution up to chain homotopy equivalence.

Definition 2.3.7 (Projective Resolutions)

Let R be a ring. Let M be an R -module. A projective resolution of M is a chain complex $P_\bullet$ of R -modules such that the following are true.

- P_n is projective for all $n \geq 0$.
- $P_\bullet$ is a left resolution of M .

Example 2.3.8

Let R be a commutative ring. An projective resolution of R in ${}_R\mathbf{Mod}$ is given by

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & 0 & \longrightarrow & R[x] & \xrightarrow{\cdot x} & R[x] \longrightarrow 0 \longrightarrow \cdots \\
 & & \downarrow & & \downarrow & & \downarrow \text{ev}_0 \\
 \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & R \longrightarrow 0 \longrightarrow \cdots
 \end{array}$$

Example 2.3.9

Let R be a commutative ring. An projective resolution of R in ${}_R\mathbf{Mod}$ is given by

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & \frac{R[x]}{(x^2)} & \xrightarrow{\cdot x} & \frac{R[x]}{(x^2)} & \xrightarrow{\cdot x} & \frac{R[x]}{(x^2)} \longrightarrow 0 \longrightarrow \cdots \\
 & & \downarrow & & \downarrow & & \downarrow \text{ev}_0 \\
 \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & R \longrightarrow 0 \longrightarrow \cdots
 \end{array}$$

Proposition 2.3.10

Let R be a ring. Let M, N be R -modules. Let $\phi : M \rightarrow N$ be an R -module homomorphism. Suppose that there are projective resolutions

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & P_2 & \longrightarrow & P_1 & \longrightarrow & M \longrightarrow 0 \\
 & & & & & & \downarrow \phi \\
 \cdots & \longrightarrow & Q_2 & \longrightarrow & Q_1 & \longrightarrow & N \longrightarrow 0
 \end{array}$$

for M and N respectively. Then there exists a chain map extending ϕ such that the following diagram commutes:

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & P_2 & \longrightarrow & P_1 & \longrightarrow & A \longrightarrow 0 \\
 & & \phi_2 \downarrow & & \phi_1 \downarrow & & \downarrow \phi \\
 \cdots & \longrightarrow & Q_2 & \longrightarrow & Q_1 & \longrightarrow & A' \longrightarrow 0
 \end{array}$$

Moreover, any two such chain maps are homotopic.

Corollary 2.3.11

Let R be a ring. Let M be an R -module. Let $P_\bullet$ and $Q_\bullet$ be projective resolutions of M . Then $P_\bullet$ and $Q_\bullet$ are chain homotopy equivalent.

Proposition 2.3.12

Let R be a ring. Let M be an R -module. Then M admits a unique projective resolution up to chain homotopy equivalence.

Definition 2.3.13 (Free Resolutions)

Let R be a ring. Let M be an R -module. A free resolution of M is a chain complex $F_\bullet$ of R -modules such that the following are true.

- F_n is free for all $n \geq 0$.
- $F_\bullet$ is a left resolution of M .

2.4 The Ext Functor**Definition 2.4.1 (The Ext Groups)**

Let R be a ring. Let M, N be R -modules. Define the n th Ext group of M and N to be

$$\text{Ext}_R^n(M, N) = H^n(\text{Hom}_R(M, I^\bullet))$$

where $I^\bullet$ is an injective resolution of N .

Example 2.4.2 Let $n, m \geq 1$. Then we have

$$\mathrm{Ext}_{\mathbb{Z}}^1\left(\frac{\mathbb{Z}}{n\mathbb{Z}}, \frac{\mathbb{Z}}{m\mathbb{Z}}\right) \cong \frac{\mathbb{Z}}{\mathrm{gcd}(m, n)\mathbb{Z}}$$

Proof Notice that a free resolution for $\mathbb{Z}/n\mathbb{Z}$ is given by

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

Applying the functor $\mathrm{Hom}(-, \mathbb{Z}/m\mathbb{Z})$ gives a chain complex of the form

$$0 \longrightarrow \mathbb{Z}/\mathrm{gcd}(m, n)\mathbb{Z} \xrightarrow{1 \mapsto \frac{m}{\mathrm{gcd}(n, m)}} \mathbb{Z}/m\mathbb{Z} \xrightarrow{\times n} \mathbb{Z}/m\mathbb{Z} \longrightarrow 0$$

Then the first cohomology group of the chain complex is precisely $\frac{\mathbb{Z}/m\mathbb{Z}}{n\mathbb{Z}/m\mathbb{Z}}$. I claim that this $\mathbb{Z}$ -module is equal to $\frac{\mathbb{Z}/m\mathbb{Z}}{\mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}}$. Let $[nk] \in \mathbb{Z}/m\mathbb{Z}$. Then $[nk] = \left[\mathrm{gcd}(n, m) \frac{nk}{\mathrm{gcd}(n, m)}\right]$ so that $[nk] \in \mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}$. Conversely, let $[\mathrm{gcd}(n, m)k] \in \mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}$. By Bezout's lemma, we know that $\mathrm{gcd}(n, m) = nx + my$ for some $x, y \in \mathbb{Z}$. Then $[\mathrm{gcd}(n, m)k] = [nmx + mky] = [nkx] \in \mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}$. Hence $[\mathrm{gcd}(n, m)k] \in n\mathbb{Z}/m\mathbb{Z}$. Thus we have $\frac{\mathbb{Z}/m\mathbb{Z}}{n\mathbb{Z}/m\mathbb{Z}} = \frac{\mathbb{Z}/m\mathbb{Z}}{\mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}}$. By the third isomorphism theorem, the first cohomology group is isomorphic to $\mathbb{Z}/\mathrm{gcd}(m, n)\mathbb{Z}$. ■

Proposition 2.4.3

Let R be a ring. Let M, N be R -modules. Then there is an isomorphism

$$\mathrm{Ext}_R^n(M, N) = H^n(\mathrm{Hom}(P^\bullet, N))$$

where $P^\bullet$ is a projective resolution of M .

Proposition 2.4.4

Let R be a ring. Let M, N be R -modules. Then the following are true.

- $\mathrm{Ext}_R^0(M, N) = \mathrm{Hom}_R(M, N)$.
- If M is projective or N is injective, then $\mathrm{Ext}_R^n(M, N) = 0$ for all $n \geq 1$.

Proposition 2.4.5

Let R be a ring. Then the following are true.

- Let M be an R -module. If $\mathrm{Ext}_R^1(M, N) = 0$ for all R -modules N , then M is projective.
- Let N be an R -module. If $\mathrm{Ext}_R^1(M, N) = 0$ for all R -modules M , then N is injective.

Proposition 2.4.6

Let R be a ring. Then the following are true.

- Suppose that $\{M_i \mid i \in I\}$ is a collection of R -modules and N is an R -module. Then there is a natural isomorphism

$$\mathrm{Ext}_R^n\left(\bigoplus_{i \in I} M_i, N\right) \cong \prod_{i \in I} \mathrm{Ext}_R^n(M_i, N)$$

- Suppose that $\{N_i \mid i \in I\}$ is a collection of R -modules and M is an R -module. Then there is a natural isomorphism

$$\mathrm{Ext}_R^n\left(M, \prod_{i \in I} N_i\right) \cong \prod_{i \in I} \mathrm{Ext}_R^n(M, N_i)$$

2.5 Flat Modules

Definition 2.5.1 (Flat Modules) Let R be a ring. An R -module M is said to be flat if for every injective linear map $\phi : K \rightarrow L$ of R -modules, the map

$$\phi \otimes \text{id}_M : K \otimes_R M \rightarrow L \otimes_R M$$

is also injective.

Proposition 2.5.2 Let R be a ring and M an R -module. Then M is a flat module if and only if for every short exact sequence $0 \rightarrow K \rightarrow L \rightarrow J \rightarrow 0$, the sequence

$$0 \rightarrow K \otimes_R M \rightarrow L \otimes_R M \rightarrow J \otimes_R M \rightarrow 0$$

is also exact.

Proposition 2.5.3 Let R be a ring. Then the following are true.

- Product: If A and B are flat over R then $A \otimes_R B$ is flat over R
- Base Change: Let S be an R -algebra ($R \rightarrow S$ a ring hom). Then $M \otimes_R S$ is flat over S for any flat R -module M
- Transitivity: Let S be an R -algebra such that S is flat over R . If C is flat over S then C is flat over R .

We have the following inclusion of modules

$$\text{Free Modules} \subset \text{Projective Modules} \subset \text{Flat Modules} \subset \text{Torsion Free Modules}$$

It is important to note that the “duality” between projective and injective modules is given from the bi-functor $\text{Hom}(-, -)$ while the “duality” given between projective and flat modules is given by the tensor-hom adjunction.

TBA: Pairing

Free implies flat

2.6 The Tor Functor

Definition 2.6.1 (The Tor Groups)

Let R be a ring. Let M, N be R -modules. Define the n th tor group of M and N to be

$$\text{Tor}_n^R(M, N) = \mathbb{H}_n(P_\bullet \otimes_R N)$$

where $P_\bullet$ is a projective resolution of M .

Proposition 2.6.2

Let R be a ring. Let M, N be R -modules. Then there is an isomorphism

$$\text{Tor}_n^{\mathbb{Z}}(M, N) \cong H_n(M \otimes_R P_\bullet)$$

where $P_\bullet$ is a projective resolution of N .

Proposition 2.6.3

Let R be a ring. Let M, N be R -modules. Then the following are true.

- $\text{Tor}_0^R(M, N) = M \otimes_R N$.
- If M or N is flat, then $\text{Tor}_n^R(M, N) = 0$ for all $n \geq 1$.

Proposition 2.6.4

Let R be a ring. Then the following are true.

- Let M be an R -module. If $\text{Tor}_1^R(M, N) = 0$ for all R -modules N , then M is flat.

- Let N be an R -module. If $\text{Tor}_1^R(M, N) = 0$ for all R -modules M , then N is flat.

Proposition 2.6.5

Let R be a ring. Then the following are true.

- Let $\{M_i \mid i \in I\}$ be a collection of R -modules. Let N be an R -module. Then there is a natural isomorphism

$$\text{Tor}_n^R\left(\bigoplus_{i \in I} M_i, N\right) \cong \bigoplus_{i \in I} \text{Tor}_n^R(M_i, N)$$

for all $n \in \mathbb{N}$.

- Let $\{N_i \mid i \in I\}$ be a collection of R -modules. Let M be an R -module. Then there is a natural isomorphism

$$\text{Tor}_n^R\left(M, \bigoplus_{i \in I} N_i\right) \cong \bigoplus_{i \in I} \text{Tor}_n^R(M, N_i)$$

for all $n \in \mathbb{N}$.

Proposition 2.6.6

Let R be a commutative ring. Let M, N be R -modules. Then there is a natural isomorphism

$$\text{Tor}_n^R(M, N) \cong \text{Tor}_n^R(N, M)$$

for all $n \in \mathbb{N}$.

2.7 The Universal Coefficient Theorems

Theorem 2.7.1 (Universal Coefficient Theorem for Cohomology) Let R be a PID. Let $(C_\bullet, \partial_\bullet)$ be a chain complex. Let M be an R -module. Then there is a split exact sequence (that does not split naturally) of the form

$$0 \longrightarrow \text{Ext}_R(H_{n-1}(C_\bullet; R), M) \longrightarrow H^n(C_\bullet; M) \xrightarrow{h} \text{Hom}(H_n(C_\bullet; R), M) \longrightarrow 0$$

where h is the map defined by $[\alpha] \mapsto [\overline{\alpha|_{Z_n(C_\bullet)}}]$ for any $n \in \mathbb{N}$.

Proof I claim that $\alpha|_{Z_n}$ descends to a map in homology. This means that there exists a unique map $\overline{\alpha|_{Z_n}} : H_n(C_\bullet) \rightarrow G$ such that the following diagram commutes

$$\begin{array}{ccc} Z_n & \xrightarrow{p} & \frac{Z_n}{B_n} = H_n(C_\bullet) \\ & \searrow \alpha|_{Z_n} & \downarrow \exists! \overline{\alpha|_{Z_n}} \\ & & G \end{array}$$

where p is the projection map. But this is true since for any $x \in C_{n+1}$ such that $\partial x \in B_n$, we have that

$$\alpha(\partial(x)) = \delta(\alpha(x)) = 0$$

where the last equality comes from the fact that $\alpha \in \ker(\delta : C^n \rightarrow C^{n+1})$. By the universal property of quotients, $\overline{\alpha|_{Z_n}}$ exists and is unique. We finally define h by

$$h([\alpha]) = [\overline{\alpha|_{Z_n}}]$$

I claim that this is well defined. This means to show that if $[\alpha] = [\beta]$ then $h([\alpha]) = h([\beta])$. So suppose that $[\alpha] = [\beta]$, then $\alpha + y = \beta$ for some $y \in B^n = \text{im}(\delta : C^{n-1} \rightarrow C^n)$. This means that

there exists $\bar{y} \in C^{n-1}$ such that $\delta(\bar{y}) = y$. Now we have that

$$\begin{aligned}
 h([\beta]) &= [\overline{\beta}|_{Z_n}] \\
 &= [\overline{(\alpha + y)}|_{Z_n}] \\
 &= [\overline{(\alpha + \delta(\bar{y}))}|_{Z_n}] \\
 &= [\overline{(\alpha + \bar{y} \circ \partial)}|_{Z_n}] \\
 &= [\overline{(\alpha)}|_{Z_n}] \quad (\partial \text{ restricted to } Z_n \text{ is just } 0) \\
 &= h([\alpha])
 \end{aligned}$$

Thus h is well defined.

We now show that h is split surjective. Define a map $v : \text{Hom}(H_n(C_\bullet, G)) \rightarrow H^n(C_\bullet, G)$ as follows. Let $\theta \in \text{Hom}(H_n(C_\bullet), G)$. Precompose with the projection map $p : Z_n(C_\bullet) \rightarrow H_n(C_\bullet)$ to obtain a map $\theta \circ p : Z_n \rightarrow G$. Now C_n being free implies that B_{n-1} is free. It follows that there is a split exact sequence

$$0 \longrightarrow Z_n \hookrightarrow C_n \xrightarrow{\partial} B_{n-1} \longrightarrow 0$$

by lmm 1.2.10. By the characterization of split exactness we conclude that there is a map $q : C_n \rightarrow Z_n$ such that the map $Z_n \hookrightarrow C_n \xrightarrow{q} Z_n$ is the identity (so q is the identity on Z_n). Now $\theta \circ p \circ q : C_n \rightarrow G$ vanishes on B_n because $\theta \circ p$ does. This means that for all $x \in C_{n+1}$, we have that $(\theta \circ p \circ q)(\partial x) = 0$ for $\partial x \in B_n$. This means that $\delta(\theta \circ p \circ q) = 0$ and $\theta \circ p \circ q \in \ker(\delta)$. Now define $v : \text{Hom}(H_n(C_\bullet, G)) \rightarrow H^n(C_\bullet, G)$ by

$$v(\theta) = [\theta \circ p \circ q]$$

Now we have that

$$\begin{aligned}
 h(v(\theta)) &= h([\theta \circ p \circ q]) \\
 &= [\overline{\theta \circ p \circ q}|_{Z_n}] \\
 &= [\overline{\theta \circ p}] \quad (q \text{ is the identity on } Z_n)
 \end{aligned}$$

Recall that $\overline{\theta \circ p}$ is the unique map such that $\overline{\theta \circ p} \circ p = \theta \circ p$ (first diagram of this proof). Since θ satisfies the description we have that $\overline{\theta \circ p} = \theta$. We conclude that $h(v(\theta)) = \theta$. From this, we have that for any $\theta \in \text{Hom}(H_n(C_\bullet), G)$ we have $v(\theta) \in H^n(C_\bullet)$ such that $h(v(\theta)) = \theta$. This proves that h is surjective. Moreover, the existence of the map v proves that the sequence

$$0 \longrightarrow \ker(h) \hookrightarrow H^n(C_\bullet) \xrightarrow{h} \text{Hom}(H_n(C_\bullet), G) \longrightarrow 0$$

is split exact.

We finally need to show the involvement of the Ext group.

We begin with a short exact sequence of chain complexes of the form

$$0 \longrightarrow Z_\bullet \hookrightarrow C_\bullet \xrightarrow{\partial} B_{\bullet-1} \longrightarrow 0$$

where its expanded form is given by

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & Z_n & \hookrightarrow & C_n & \xrightarrow{\partial} & B_{n-1} \longrightarrow 0 \\
 & & \downarrow 0 & & \downarrow \partial & & \downarrow 0 \\
 0 & \longrightarrow & Z_{n-1} & \hookrightarrow & C_{n-1} & \xrightarrow{\partial} & B_{n-2} \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

Dualizing the entire diagram gives

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \uparrow & & \uparrow & & \uparrow & \\
 0 & \longleftarrow & \text{Hom}(Z_n, G) & \longleftrightarrow & \text{Hom}(C_n, G) & \xleftarrow{\partial^*} & \text{Hom}(B_{n-1}, G) \longrightarrow 0 \\
 & \uparrow 0 & & \uparrow \partial^* & & \uparrow 0 & \\
 0 & \longleftarrow & \text{Hom}(Z_{n-1}, G) & \longleftrightarrow & \text{Hom}(C_{n-1}, G) & \xleftarrow{\partial^*} & \text{Hom}(B_{n-2}, G) \longrightarrow 0 \\
 & \uparrow & & \uparrow & & \uparrow & \\
 & \vdots & & \vdots & & \vdots &
 \end{array}$$

Now each row is split exact because it is split exact before dualizing and the Hom functor preserves split exact sequences. We think of the above as a short exact sequence of chain complexes. This induces a long exact sequence in homology given by

$$\cdots \longrightarrow H_{n-1}(\text{Hom}(Z_{n-1}, G)) \longrightarrow H_n(\text{Hom}(B_{n+1}, G)) \longrightarrow H^n(C_\bullet, G) \longrightarrow H_n(\text{Hom}(Z_n, G)) \longrightarrow H_{n-1}(\text{Hom}(B_n, G)) \longrightarrow \cdots$$

Since $\text{Hom}(Z_\bullet, G)$ and $\text{Hom}(B_\bullet, G)$ have 0 as the boundary map, the above long exact sequence reduces to

$$\cdots \longrightarrow \text{Hom}(Z_{n-1}, G) \longrightarrow \text{Hom}(B_{n-1}, G) \longrightarrow H^n(C_\bullet, G) \longrightarrow \text{Hom}(Z_n, G) \longrightarrow \text{Hom}(B_n, G) \longrightarrow \cdots$$

Let us work out the connecting homomorphism $\text{Hom}(Z_n, G) \rightarrow \text{Hom}(B_{n-1}, G)$. For $\alpha \in \text{Hom}(Z_n, G)$, the connecting homomorphism sends $\alpha : Z_n \rightarrow G$ to $\gamma : B_{n-1} \rightarrow G$ where $\partial(\beta) = \alpha$ and $i_n(\gamma) = \partial(\beta)$. Now notice that

$$i_n(\gamma) = \partial(\beta) = \alpha$$

for $i_n : B_{n-1} \rightarrow C_n$ the inclusion. Thus the connecting homomorphism is precisely the dual of i_n , denoted i_n^* .

I claim that there is an isomorphism

$$\ker(i_n^* : \text{Hom}(Z_n, G) \rightarrow \text{Hom}(B_n, G)) \cong \text{Hom}(H_n(C_\bullet), G)$$

Let $\alpha \in \ker(i_n^*)$. Then $\alpha : Z_n \rightarrow G$ is such that $\alpha \circ i_n : B_n \hookrightarrow Z_n \rightarrow G$ is the zero map. Or in other words, $\alpha|_{B_n} = 0$. But then α is clearly an element of $\text{Hom}(H_n(C_\bullet), G)$ by definition. The converse is also true.

Now by definition of $H^n(C_\bullet)$, we get an exact sequence

$$0 \longrightarrow \ker(h) \hookrightarrow H^n(C_\bullet) \xrightarrow{h} \text{Hom}(H_n(C_\bullet), G) \longrightarrow 0$$

Going back to the long exact sequence

$$\cdots \longrightarrow \text{Hom}(Z_{n-1}, G) \xrightarrow{i_{n-1}^*} \text{Hom}(B_{n-1}, G) \longrightarrow H^n(C_\bullet, G) \longrightarrow \text{Hom}(Z_n, G) \xrightarrow{i_n^*} \text{Hom}(B_n, G) \longrightarrow \cdots$$

We can restrict the $\text{Hom}(B_{n-1}, G)$ term to $\text{coker}(i_{n-1}^*)$ and the $\text{Hom}(Z_n, G)$ term to $\ker(i_n^*)$ to obtain a short exact sequence

$$0 \xrightarrow{i_{n-1}^*} \text{coker}(i_{n-1}^*) \longrightarrow H^n(C_\bullet, G) \longrightarrow \ker(i_n^*) \xrightarrow{i_n^*} 0$$

Now using the isomorphism $\ker(i_n^*) \cong \text{Hom}(H_n(C_\bullet), G)$ it is also easy to see that the map $H^n(C_\bullet, G) \rightarrow \ker(i_n^*)$ is actually just the map h . We conclude that the above long exact sequence is precisely the long exact sequence we wanted to analyse, and in particular $\text{coker}(i_{n-1}^*) \cong \ker(h)$.

Now consider the following exact sequence:

$$0 \longrightarrow B_{n-1} \xrightarrow{i_{n-1}} Z_{n-1} \twoheadrightarrow H_{n-1}(C_\bullet) \longrightarrow 0$$

Since C_n is free abelian we conclude that B_{n-1} and Z_{n-1} are free. This means that this exact sequence is actually a free resolution of $H_{n-1}(C_\bullet)$. Using this free resolution, we obtain an isomorphism

$$\text{Ext}(H_{n-1}(C_\bullet), G) \cong \text{coker}(i_{n-1}^*)$$

Thus we conclude the proof. ■

Corollary 2.7.2 Let $(C_\bullet, \partial_\bullet)$ be a chain complex of free abelian groups. Let k be a field. Then there is an isomorphism

$$H^n(C_\bullet, k) \cong \text{Hom}(H_n(C_\bullet), k)$$

given by the universal coefficient theorem.

Theorem 2.7.3 (Universal Coefficient Theorem for Homology) Let $C_\bullet$ be a chain complex of free abelian groups. Let A be an abelian group. Then there exists a natural map $h : H_n(C_\bullet) \otimes A \rightarrow H_n(C_\bullet; A)$ such that $\text{coker}(h) \cong \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(C_\bullet), A)$ and a split exact sequence (that is not natural) of the form

$$0 \longrightarrow H_n(C_\bullet) \otimes A \xrightarrow{h} H_n(C_\bullet; A) \longrightarrow \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(C_\bullet), A) \longrightarrow 0$$

for any $n \in \mathbb{N}$. In particular, split exactness implies that there is an isomorphism

$$H_n(C_\bullet; A) \cong H_n(C_\bullet) \otimes A \oplus \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(C_\bullet), A)$$

for any $n \in \mathbb{N}$.

2.8 Extensions and the Ext Groups

Definition 2.8.1 (Extensions) Let R be a ring. Let M, N be R -modules. An extension of M by N is a short exact sequence

$$0 \longrightarrow N \longrightarrow E \longrightarrow M \longrightarrow 0$$

of R -modules.

Definition 2.8.2 (Equivalent Extensions) Let R be a ring. Let M, N be R -modules. We say that two extensions

$$0 \longrightarrow N \longrightarrow E \longrightarrow M \longrightarrow 0$$

$$0 \longrightarrow N \longrightarrow F \longrightarrow M \longrightarrow 0$$

are equivalent if there exists an R -module homomorphism $\phi : E \rightarrow F$ such that the following diagram commutes:

$$\begin{array}{ccccccc} 0 & \longrightarrow & N & \longrightarrow & E & \longrightarrow & M \longrightarrow 0 \\ & & \downarrow \text{id}_N & & \downarrow \phi & & \downarrow \text{id}_M \\ 0 & \longrightarrow & N & \longrightarrow & F & \longrightarrow & M \longrightarrow 0 \end{array}$$

Proposition 2.8.3 Let R be a ring. Let M, N be R -modules. Suppose that the following two extensions are equivalent:

$$\begin{array}{ccccccc} 0 & \longrightarrow & N & \longrightarrow & E & \longrightarrow & M \longrightarrow 0 \\ & & \downarrow \text{id}_N & & \downarrow \phi & & \downarrow \text{id}_M \\ 0 & \longrightarrow & N & \longrightarrow & F & \longrightarrow & M \longrightarrow 0 \end{array}$$

Then ϕ is an isomorphism. Moreover, equivalent extensions is an equivalence relation.

Definition 2.8.4 (Split Extensions) Let R be a ring. Let M, N be R -modules. We say that an extension splits if it is equivalent to the following extension

$$0 \longrightarrow N \longrightarrow N \oplus M \longrightarrow M \longrightarrow 0$$

of R -modules. The above extension is called a trivial extension.

Proposition 2.8.5 Let R be a ring. Let M, N be R -modules. There is a bijection

$$\frac{\{\text{Extensions of } M \text{ by } N\}}{\cong} \xleftrightarrow{1:1} \text{Ext}_R^1(M, N)$$

where $\cong$ means equivalence of extensions. Moreover, the trivial extension corresponds to the zero element of $\text{Ext}_R^1(M, N)$.

Proposition 2.8.6 Let A, B be R -modules. Then the following are true regarding the Ext group.

- $\text{Ext}_R^0(A, B) \cong \text{Hom}_R(A, B)$
- $\text{Ext}_R^i(A, B) = 0$ for all $i > 0$ if A is projective or B is injective
- $\text{Ext}_R^i(A, B) = 0$ for all $i \geq 2$ if A, B are $\mathbb{Z}$ -modules.

TBA: External products

3 Homological Algebra of Abelian Groups

3.1 Ext and Tor for Abelian Groups

Lemma 3.1.1

Let A be an abelian group. Then the following is a free resolution of A :

$$0 \longrightarrow \ker(\pi) \longrightarrow \bigoplus_{a \in A} \mathbb{Z} \cdot a \xrightarrow{\pi} A \longrightarrow 0$$

where $\pi : \bigoplus_{a \in A} \mathbb{Z} \cdot a \rightarrow A$ is the projection map.

Proof Clearly the sequence is exact. Moreover, every subgroup of a free abelian group is free abelian. Hence $\ker(\pi)$ is free abelian and the sequence is indeed a free resolution. ■

Proposition 3.1.2

Let A, B be abelian groups. Then we have

$$\operatorname{Ext}_{\mathbb{Z}}^n(A, B) = 0$$

for all $n \geq 2$.

Proposition 3.1.3

Let A be an abelian group. Then the following are true.

- $\operatorname{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/n\mathbb{Z}, A) \cong A/nA$.
- $\operatorname{Ext}_{\mathbb{Z}}^1(A, \mathbb{Q}) = 0$.

Proposition 3.1.4

Let A, B be abelian groups. Then we have

$$\operatorname{Tor}_n^{\mathbb{Z}}(A, B) = 0$$

for all $n \geq 2$.

Proposition 3.1.5

Let A, B be finitely generated abelian groups. Then we have

$$\operatorname{Tor}_1^{\mathbb{Z}}(A, B) \cong A_{\text{tor}} \otimes_{\mathbb{Z}} B_{\text{tor}}$$

3.2 Divisible Groups

(can be removed to Infinite Abelian Groups)

Definition 3.2.1 (Divisible Groups) Let A be an abelian group. Let $n \in \mathbb{N} \setminus \{0\}$. We say that A is n -divisible if for all $g \in A$, there exists $h \in A$ such that $nh = g$.

Proposition 3.2.2 Let A be an abelian group. Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are equivalent.

- A is n -divisible.
- $nA = A$.
- A is an injective $\mathbb{Z}$ -module.

Definition 3.2.3 (The Prüfer p -Group) Let p be a prime. Define the Prüfer p -Group to be the quotient group

$$\mathbb{Z}(p^\infty) = \frac{\mathbb{Z}[1/p]}{\mathbb{Z}}$$

Example 3.2.4 The following are divisible groups.

- $\mathbb{Q}$.
- $\mathbb{Z}[1/p]/\mathbb{Z}$.

Theorem 3.2.5 (Structure Theorem for Divisible Groups) Let A be an abelian group. If A is divisible, then there is an isomorphism

$$A \cong \left(\bigoplus_{p \text{ is prime}} \frac{\mathbb{Z}[1/p]}{\mathbb{Z}}^{k_p} \right) \oplus \mathbb{Q}^I$$

for some choice of indexing sets I and some $k_p \in \mathbb{N}$.

4 Graded Ext and Tor Groups

4.1 Graded Free Modules

Definition 4.1.1 (Graded Free Modules)

Let R be a graded ring. Let M be a graded R -module. We say that M is a graded free R -module if M admits a basis consisting of homogeneous elements.

Lemma 4.1.2

Let R be a graded ring. Let M be a graded R -module. Then M is a graded free R -module if and only if there is a graded isomorphism

$$M \cong \bigoplus_{i \in I} R(-d_i)$$

for some $d_i \in \mathbb{N}$.

4.2 Graded Injective Modules

Definition 4.2.1 (Injective Modules)

Let R be a graded ring. Let I be a graded R -module. We say that I is injective if for all short exact sequences of graded R -modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

we have that

$$0 \longrightarrow \operatorname{Hom}_R(M_3, I) \longrightarrow \operatorname{Hom}_R(M_2, I) \longrightarrow \operatorname{Hom}_R(M_1, I) \longrightarrow 0$$

is an exact sequence of graded R -modules.

Proposition 4.2.2

Let R be a graded ring. Let I be a graded R -module. Then I is injective if and only if for graded R -modules M, N and all injective graded homomorphisms $f : M \rightarrow N$ and all graded maps $u : M \rightarrow I$, there exists a graded morphism $N \rightarrow I$ such that the following diagram commutes:

$$\begin{array}{ccc} M & \xrightarrow{u} & I \\ f \downarrow & \nearrow \exists! & \\ N & & \end{array}$$

Example 4.2.3

Let k be a field. Let $R = k[x]$ with $\deg(x) = 1$. Then $k[x, x^{-1}]/k[x]$ is an injective graded R/xR -module, but it is not injective as an ungraded module.

4.3 Projective Modules

Definition 4.3.1 (Projective Modules)

Let R be a graded ring. Let P be a graded R -module. We say that P is projective if for all short exact sequences of graded R -modules

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

we have that

$$0 \longrightarrow \operatorname{Hom}_R(P, M_1) \longrightarrow \operatorname{Hom}_R(P, M_2) \longrightarrow \operatorname{Hom}_R(P, M_3) \longrightarrow 0$$

is an exact sequence of graded R -modules.

Lemma 4.3.2

Let R be a graded ring. Let F be a graded free R -module. Then F is projective.

Proposition 4.3.3

Let R be a graded ring. Let P be a graded R -module. Then the following are equivalent.

- P is projective.
- For all graded R -modules M, N and all surjective graded homomorphisms $f : M \rightarrow N$ and graded $u : P \rightarrow N$, there exists a graded morphism $P \rightarrow M$ such that the following diagram commutes:

$$\begin{array}{ccc} & M & \\ \exists! \nearrow & \downarrow f & \\ P & \xrightarrow{u} & N \end{array}$$

- $P \oplus Q$ is a graded free R -module for some graded R -module Q .

Proposition 4.3.4

Let R be a graded ring. Let P be a graded R -module that is projective. Then P is projective as an ungraded R -module.